

CARS EDA PROJECT DOCUMENT:

- Performed an end-to-end exploratory data analysis on a used-car dataset using Python, Pandas, NumPy, Matplotlib and Seaborn.
- Analyzed **5,961 records and 15 columns** covering vehicle details, usage, specifications, ownership and pricing.
- Performed data-quality checks for shape, columns, data types, descriptive statistics, missing values and duplicate records.
- Converted **Engine, Power and Mileage** from text-formatted values into numeric fields and created a derived **Brand** feature.
- Used countplots, histograms/KDE, scatterplots, boxplots and a correlation heatmap for univariate and bivariate analysis.
- Identified the strongest numerical relationships with Price in the notebook, including **Power (0.773)** and **Engine (0.659)**.

Tools & Technology used: Python, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook

Why this project ?

The notebook studies used-car pricing by examining vehicle age, kilometers driven, mileage, engine, power, seats, transmission, fuel type, owner type, location and brand in relation to the target variable **Price**.

Dataset overview

Initial shape: **5,961 rows × 15 columns**. One duplicate was found and removed, leaving 5,960 rows for the later Price analysis. The data contains 212 unique car names, 11 locations, 5 fuel types, 2 transmission types and 4 owner types. **New_Price** has 5,137 missing values (about **86.18%**), the largest missing-value count.

Features ?

- 1 Name: car name/brand and model
- 2 Location: selling/availability city
- 3 Year: manufacturing year
- 4 Kilometers_Driven: total kilometers driven
- 5 Fuel_Type: fuel category
- 6 Transmission: Automatic or Manual
- 7 Owner_Type: ownership category
- 8 Mileage: mileage value
- 9 Engine: engine displacement in CC
- 10 Power: maximum engine power in bhp
- 11 Colour: vehicle colour
- 12 Seats: number of seats
- 13 No. of Doors: number of doors
- 14 New_Price: new-car price where available
- 15 Price: used-car price

Problem?

- 1 Which vehicle and usage variables are associated with used-car Price?
- 2 How do Year and Kilometers_Driven relate to Price?
- 3 How do Mileage, Engine and Power relate to Price?
- 4 How do Transmission, Fuel_Type and Owner_Type relate to Price?
- 5 How do Location and Brand differ in average Price?
- 6 How are numerical variables correlated with Price and with each other?

Analysis performed

- **Price:** descriptive statistics, histogram/KDE and boxplot. Mean $\approx$ **9.53**; minimum 0.44; maximum 160.
- **Univariate:** countplots for Fuel_Type, Transmission and Owner_Type; histogram/KDE plots for numerical variables.
- **Bivariate:** scatterplots for Year, Kilometers_Driven, Mileage, Engine and Power versus Price; boxplots for Transmission, Fuel_Type, Owner_Type and Seats.
- **Location:** mean Price calculated by location and displayed as a bar chart.
- **Brand:** mean Price calculated for each derived Brand.
- **Correlation:** numerical correlation matrix and heatmap created after converting Mileage to numeric form.
- **Automated plotting:** loops generate countplots for categorical columns and histogram/KDE plots for numerical columns.

Key findings from the notebook

- 1 Price correlation: **Power 0.773, Engine 0.659, Mileage -0.313, Year 0.305, and Kilometers_Driven -0.011.**
- 2 Engine and Power have a strong correlation of approximately **0.867**.
- 3 Mean Price by location is highest in the calculated list for **Coimbatore (15.11)**, followed by Bangalore (13.50) and Kochi (11.22).
- 4 Mean Price varies substantially by Brand; the calculated list includes Lamborghini (120.00), Bentley (59.00) and Porsche (48.35) at the top.
- 5 New_Price contains substantial missing data, while most other columns have comparatively small missing-value counts.

Important points

- 1 The notebook shows associations between used-car Price and several vehicle specifications, especially Power and Engine in the computed correlation matrix.
- 2 Location and Brand have different average Price levels; the notebook does not establish causation.
- 3 The high proportion of missing New_Price values should be considered in any further analysis or modeling.
- 4 Further work can include systematic missing-value treatment, feature engineering and predictive modeling.